

Curriculum Vitae – Hao Yan
Associate Professor
School of Computing and Augmented Intelligence
Arizona State University, Tempe
Phone: (480) 727-0556 Email: haoyan@asu.edu
Website: <https://faculty.engineering.asu.edu/hao-yan/>

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I. EDUCATION

- Ph.D.** Industrial Engineering, Georgia Institute of Technology, 2017, Minor: Machine Learning
Ph.D. advisors: Dr. Jianjun Shi and Dr. Kamran Paynabar
- M.S.** Statistics, Georgia Institute of Technology, 2015
- M.S.** Computational Science and Engineering, 2016
- B.S.** Economics, Peking University, 2011
- B.S.** Physics, Peking University, 2011

II. Professional Experiences

Arizona State University, Tempe, AZ

Associate Professor, August 2024 – Present

- Faculty member in the School of Computing and Augmented Intelligence

Assistant Professor, August 2017 – August 2024

- Faculty member in the School of Computing and Augmented Intelligence

Graduate Research Assistant Jan 2012 – Aug 2017

- Lab manager and research assistant in the System Informatics and Control Lab in Georgia Institution of Technology

Teaching Assistant, 2012 – 2014

- Teaching assistant for a graduate-level course in data science and statistics

III. Honors, Awards, And Recognitions

A. NATIONAL SOCIETIES AWARDS/HONORABLE MENTIONS

- A1. IEEE International Conference on Automation Science and Engineering (CASE) Best Conference Paper Award, 2025, for the paper “Multi-modal Generative Modeling of Event Sequences and Time Series for Solar PV Systems” (see **C6**).
- A2. International Conference on Quality, Statistics, and Reliability (ICQSR) Best Paper Award Finalist, 2025, for the paper “Exact Multistage Bayesian Optimization”.
- A3. INFORMS Quality, Statistics, and Reliability (QSR) Section Best Refereed Paper Competition Finalist, 2024, for the paper “Multi-physics Guided Generative Diffusion Models with Manufacturing Applications”.
- A4. IIEE DAIS Best Track Paper Competition, 2023, for the paper “Interpretation and visualization of distance covariance with the additive decomposition of correlations formula” (see **J4**).
- A5. Institute of Industrial and Systems Engineers (IIEE) Data Analytics and Information Systems (DAIS) Division Teaching Award, 2022
- A6. IIEE Quality Control & Reliability Engineering (QCRE) Best Track Paper Competition Finalist, 2023, for the paper “Graph-aware Tensor Topic Models for Individualized Passenger Travel Pattern Clustering”
- A7. 16th Institute for Operations Research and the Management Science (INFORMS) Hybrid Workshop on Data Mining and Decision Analytics theoretical track finalist, 2021, for the paper “Individualized Passenger Travel Pattern Multi-clustering based on Graph Regularized Tensor Latent Dirichlet Allocation” (see **J34**).
- A8. Institute of Electrical and Electronics Engineers (IEEE) Conference on Automation Science and Engineering (CASE) Best Paper Award, 2020, for the paper “Long-Short Term Spatiotemporal Tensor Prediction for Passenger Flow Profile” (see **J49**). The chosen paper is selected among all papers submitted to IEEE CASE 2020.
- A9. IIEE Transactions Best Paper Award with Focus Issue on Quality and Reliability Engineering, 2019 for the paper “Weakly correlated profile monitoring based on sparse multi-channel functional principal component analysis” (see **J56**). The chosen paper is selected from among all publications submitted to IIEE Transaction in Year 2019.
- A10. American Society of Quality (ASQ) Brumbaugh Award, 2019, for the paper “Multiple Profiles Sensor-Based Monitoring and Anomaly Detection” (see **J54**), *Journal of Quality Technology*. This award has been presented since 1949, is given to “the paper making the *largest single contribution to the development of industrial application of quality control.*” The chosen paper is selected from among publications in the seven journals published by ASQ in a given year.
- A11. IEEE Transactions on Automation Science and Engineering Best Paper Award, for the paper “Generalized Wavelet Shrinkage of Inline Raman Spectroscopy for Quality Monitoring of Continuous Manufacturing of Carbon Nanotube Buckypaper” (see **J59**).

- A12. Best Paper Award in Data Mining Section of INFORMS (2018, Theoretical Track) for the paper “Multiple Tensor-on-Tensor Regression: An Approach for Modeling Processes with Heterogeneous Sources of Data” (see **J50**).
- A13. Best Paper Award in Data Mining Section of INFORMS (2017, Applied Track) for the paper “Dynamic Multivariate Functional Data Modeling via Sparse Subspace Learning,” (see **J48**)
- A14. Best Student Paper Award Finalist in Quality, Statistics, and Reliability (QSR) Section of INFORMS, (2016), for the paper “AKM2D: An Adaptive Framework for Online Sensing And Anomaly Detection” (see **J52**)
- A15. Best Paper Award Winner in the QSR Referred Track of INFORMS for the paper “Real-time Monitoring and Diagnosis of High-Dimensional Data Streams via Spatio-Temporal Smooth Sparse Decomposition” (see **J55**), Oct 2015.
- A16. Best Student Paper Award Winner in the IISE Quality Control and Reliability Engineering (QCRE) division, for the paper “Monitoring and Diagnostics of Streaming Images Via Recursive Smooth-Sparse Decomposition,” May 2015.
- A17. Best Student Paper Award Winner in the Data Mining Section of INFORMS, for the paper “Anomaly Detection in Images with Smooth Background Via Smooth-Sparse Decomposition” (see **J58**), Nov. 2014.

B. AWARDS/HONORABLE MENTIONS FOR ADVISED STUDENTS

- A18. IISE QCRE Best Student Paper Competition finalist, May 2022, for my Ph.D. student **Ji-uyun Hu’s** “Tensor Completion for High-dimensional Data with Graph-structured and Weakly-dependent Sample Dimension” (see **J63**).
- A19. 16th INFORMS Hybrid Workshop on Data Mining and Decision Analytics applied track Runner-up (2nd place), Sep 2021, for my Ph.D. student **Dorukhan Sergin’s** paper “Low-rank robust subspace tensor clustering for metro passenger flow modeling” (see **J14**).
- A20. INFORMS QSR Best Referred Paper Award, Oct 2020, for my Ph.D. student **Xinyu Zhao’s** paper “Adaptive Partially-Observed Sequential Change Detection and Isolation” (see **J29**).

IV. PUBLICATIONS, INTELLECTUAL PROPERTY, AND PRESENTATIONS

A1. Published and Accepted Journal Papers

- J1. Erly, S., Yan, H.*, Holte, S., Mei, Y., Kerani, R.P., 2026, “Evaluating Time-Space Methodologies to Detect Clusters of HIV Transmission: A Comparison of Advanced Methods in Washington State, 2010–2022”, *Journal of Acquired Immune Deficiency Syndromes*, 101(9), 957-963.
- J2. **Lee, Y.**, Chen, R.Q., Yan, H.*, Mupparapu, M., Lure, F., Li, J., Setzer, F.C., 2026, “Dentin-Normalized CBCT Radiomics within a Periapical Lesion Differential Diagnosis System for Differentiating Granulomas and Radicular Cysts”, *Journal of Endodontics*.
- J3. **Rathnakumar, R.**, Lu, X., Huang, J., Yan, H.*, Liu, Y., 2026, “Bayesian Entropy Neural Networks for physics-aware prediction”, *Reliability Engineering & System Safety*, 266, 111575.
- J4. **Wang, A.**, Yan, H.*, Du, J., 2026, “Interpretation and visualization of distance covariance with additive decomposition of correlations formula”, *INFORMS Journal on Data Science*, 2026. DOI: 10.1287/ijds.2024.0054.
- J5. **Kang, Y.**, Zhao, X., Feng, S., Yan, H.*, 2026, “Optimizing Multiple Condition Policy based on Real-time Degradation Signals via Model-based Reinforcement Learning”, *Journal of Quality Technology*.
- J6. **Huang, J.**, Li, J., Setzer, F.C., Lure, F., Ju, F., Yan, H.*, 2026, “Aleatoric-Aware Posterior Regularized Bayesian Neural Networks for Incorporating Domain Knowledge: Addressing Epistemic and Aleatoric Uncertainties”, *INFORMS Journal on Data Science*, accepted.
- J7. **Xu, B.**, Chen, S., Yan, H.*, 2026, “Retimed Bellman Flows: Escaping the Impossible Triangle of Velocity Bootstrapping”. *arXiv preprint arXiv:2609.32828*.
- J8. **Xie, X.**, Yan, H.*, 2026, “Personalized Federated Hierarchical Gaussian Processes for Privacy-Preserving Modeling of Heterogeneous Distributed Systems”. *arXiv preprint arXiv:2609.19337*.
- J9. Yan, H.*, Sarabi, A., Zou, Q., Xu, B., 2026, “General Value Functions for Remaining Useful Life and Failure-Mode Prediction”. *arXiv preprint arXiv:2607.22268*.
- J10. **Moradi, M.**, Chen, S., Yan, H.*, Paynabar, K., 2025, “A Single Image Is All You Need: Zero-Shot Anomaly Localization Without Training Data”. *arXiv preprint arXiv:2509.18354*.
- J11. **Chen, K.**, Xie, X., German, P., Yan, H.*, Wang, A., 2025, “Bayesian Optimization for Reactor Design Optimization”, *Transactions of the American Nuclear Society*.
- J12. Shi, N., Yan, H.*, Guo, S., Al Kontar, R., 2025, “Diffusion-Based Surrogate Modeling and Multi-Fidelity Calibration”, *IEEE Transactions on Automation Science and Engineering*.
- J13. **Zhao, X.**, Yan, H.*, Liu, Y., 2025, “Hierarchical Multilabel Classification for Fine-Level Event Extraction from Aviation Accident Reports”, *INFORMS Journal on Data Science*, 4(1), 51-66.

- J14. **Sergin, N. D.**, Hu, J., Li, Z., Zhang, C., Tsung, F., Yan, H.*, 2025, “Low-rank robust subspace tensor clustering for metro passenger flow modeling”, *INFORMS Journal on Data Science*, 4(1), 33-50.
- J15. **Dhulipala, S.L.**, German, P., Che, Y., Prince, Z.M., Xie, X., Simon, P.A., Labouré, V.M., Yan, H.*, 2025, “MOOSE ProbML: Parallelized probabilistic machine learning and uncertainty quantification for computational energy applications”, *Journal of Computational Science*, 102776.
- J16. **Lee, Y.**, Kwak, M.G., Chen, R.Q., Yan, H.*, Mupparapu, M., Lure, F., Setzer, F.C., Li, J., 2025, “Oral-Anatomical Knowledge-Informed Semi-Supervised Learning for 3D Dental CBCT Segmentation and Lesion Detection”, *IEEE Transactions on Automation Science and Engineering*.
- J17. **Xu, B.**, Kang, Y., Zhao, X., Yan, H.*, Ju, F., 2025, “Partially observable Markov decision process framework for operating condition optimization using real-time degradation signals”, *Journal of Quality Technology*, 1-26.
- J18. **Hu, J.**, Shi, N., Al Kontar, R., Yan, H.*, 2025, “Personalized tucker decomposition: Modeling commonality and peculiarity on tensor data”, *Technometrics*, 1-17.
- J19. **Sergin, N. D.**, Huang, J., Chang, T., Yan, H.*, 2024, “Image-based novel fault detection with deep learning classifiers using hierarchical labels”, *IIEE Transactions*, 56(10), 1112-1130.
- J20. **Chen, R.Q.**, Lee, Y., Yan, H.*, Mupparapu, M., Lure, F., Li, J., Setzer, F.C., 2024, “Leveraging Pretrained Transformers for Efficient Segmentation and Lesion Detection in Cone-Beam Computed Tomography Scans”, *Journal of Endodontics*, 50(10), 1505-1514.
- J21. Shi, N., Yan, H.*, Guo, S., Al Kontar, R., 2024, “Multi-physics Simulation Guided Generative Diffusion Models with Applications in Fluid and Heat Dynamics”. *arXiv preprint arXiv:2407.17720*.
- J22. Zhang, Q., Yan, H.*, Liu, Y., 2024, “Power generation forecasting for solar plants based on Dynamic Bayesian networks by fusing multi-source information”, *Renewable and Sustainable Energy Reviews*, 202, 114691.
- J23. Guo, J., Yan, H.*, Zhang, C., 2024, “Thompson Sampling-Based Partially Observable Online Change Detection for Exponential Families”, *INFORMS Journal on Data Science*, 3(2), 145-161.
- J24. **Huang, J.**, Farpour, N., Yang, B.J., Mupparapu, M., Lure, F., Li, J., Yan, H.*, Setzer, F.C., 2024, “Uncertainty-based Active Learning by Bayesian U-Net for Multi-label Cone-Beam CT Segmentation”, *Journal of Endodontics*, 50(2), 220-228.
- J25. Guo, J., Yan, H.*, Zhang, C., 2023, “A Bayesian Partially Observable Online Change Detection Approach with Thompson Sampling”, *Technometrics*, 65(2), 179-191. (**Win 2021 INFORMS Data Mining Section Best Paper Competition Finalist**).

- J26. **Biehler, M.**, Yan, H.*, Shi, J., 2023, “ANTLER: Bayesian Nonlinear Tensor Learning and Modeler for Unstructured, Varying-Size Point Cloud Data”, *IEEE Transactions on Automation Science and Engineering*, 21(1), 402-415.
- J27. **Huang, J.**, **Pang, Y.**, Liu, Y., Yan, H.*, 2023, “Posterior Regularized Bayesian Neural Network Incorporating Soft and Hard Knowledge Constraints”, *Knowledge-Based Systems*, 110043.
- J28. Wu, T., Wang, R., Yan, H.*, Shao, X., 2022, “Adaptive Change Point Monitoring for High-Dimensional Data”, *Statistica Sinica*, 32, 1583-1610.
- J29. **Zhao, X.**, Hu, J., Mei, Y., Yan, H.*, 2022, “Adaptive Partially-Observed Sequential Change Detection and Isolation”, *Technometrics*, 64(4), 502-512. (**Win 2020 INFORMS QSR Best Referred Paper Award**).
- J30. **Hu, J.**, Mei, Y., Holte, S., Yan, H.*, 2023, “Adaptive Resources Allocation CUSUM for Binomial Count Data Monitoring with Application to COVID-19 Hotspot Detection”, *Journal of Applied Statistics*, 50(14), 2889-2913.
- J31. **Pang, Y.**, Zhao, X., Hu, J., Yan, H.*, Liu, Y., 2022, “Bayesian Spatio-Temporal Graph Transformer Network (B-Star) for Multi-Aircraft Trajectory Prediction”, *Knowledge-Based Systems*, 249, 108998.
- J32. **Lahoti, G.**, Ranjan, C., Chen, J., Yan, H.*, Zhang, C., 2022, “Convolutional Neural Network-Assisted Adaptive Sampling for Sparse Feature Detection in Image and Video Data”, *IEEE Intelligent Systems*, 38(1), 45-57.
- J33. **Zhao, X.**, Yan, H.*, Hu, Z., Du, D., 2022, “Deep Spatio-temporal Sparse Decomposition for Trend Prediction and Anomaly Detection in Cardiac Electrical Conduction”, *IIEE Transactions on Healthcare Systems Engineering*, 12(2), 150-164.
- J34. **Li, Z.**, Yan, H.*, Zhang, C., Tsung, F., 2022, “Individualized Passenger Travel Pattern Multi-clustering based on Graph Regularized Tensor Latent Dirichlet Allocation”, *Data Mining and Knowledge Discovery*, 36(4), 1247-1278. (**Win 2021 INFORMS Data Mining and Decision Analytics theoretical track finalist**).
- J35. **Li, Y.**, Yan, H.*, Jin, R., 2022, “Multi-Task Learning with Latent Variation Decomposition for Multivariate Responses in a Manufacturing Network”, *IEEE Transactions on Automation Science and Engineering*, 20(1), 285-295.
- J36. **Zhao, Y.**, Yan, H.*, Holte, S., Mei, Y., 2022, “Rapid Detection of Hot-spots via Tensor Decomposition with Applications to Crime Rate Data”, *Journal of Applied Statistics*, 49(7), 1636-1662.
- J37. Yan, H.*, Grasso, M., Paynabar, K., Colosimo, B.M., 2022, “Real-time detection of clustered events in video-imaging data with applications to additive manufacturing”, *IIEE Transactions*, 54(5), 464-480.
- J38. **Li, Z.**, Yan, H.*, Tsung, F., Zhang, K., 2022, “Profile Decomposition Based Hybrid Transfer Learning for Cold-Start Data Anomaly Detection”, *ACM Transactions on Knowledge Discovery from Data*, 16(6), 1-28.

- J39. **Du, J.**, Yan, H.*, Chang, T., Shi, J., 2021, “A Tensor Voting Based Surface Anomaly Classification Approach by Using 3D Point Cloud Data”, *Journal of Manufacturing Science and Engineering*, 144(5), 051005.
- J40. **Pang, Y.***, Zhao, X., Yan, H.*, Liu, Y., 2021, “Data-Driven Trajectory Prediction with Weather Uncertainties: A Bayesian Deep Learning Approach”, *Transportation Research Part C: Emerging Technologies*, 130, 103326.
- J41. Yan, H.*, Sergin, N. D., Brenneman, W., Lange, S., Ba, S., 2021, “Deep Multistage Multi-Task Learning for Quality Prediction of Multistage Manufacturing Systems”, *Journal of Quality Technology*, 53(5), 526-544.
- J42. **Lahoti, G.**, Chen, J., Yue, X., Yan, H.*, Ranjan, C., Qian, Z., Zhang, C., Wang, B., 2021, “Image Decomposition-based Sparse Extreme Pixel-level Feature Detection with Application to Medical Images”, *IIEE Transactions on Healthcare Systems Engineering*, 11(4), 338-354.
- J43. **Fang, X.**, Yan, H.*, Gebraeel, N., Paynabar, K., 2021, “Multi-Sensor Prognostics Modeling for Applications with Highly Incomplete Signals”, *IIEE Transactions*, 53(5), 597-613.
- J44. **Sergin, N. D.**, Yan, H.*, 2021, “Toward a better monitoring statistic for profile monitoring via variational autoencoders”, *Journal of Quality Technology*, 53(5), 454-473.
- J45. Zheng, Z., Yan, H.*, Setzer, F.C., Shi, K., Mupparapu, M., Li, J., 2020, “Anatomically Constrained Deep Learning for Automating Dental CBCT Segmentation and Lesion Detection”, *IEEE Transactions on Automation Science and Engineering*, 18(2), 603-614.
- J46. Setzer, F.C., Shi, K., Zheng, Z., Yan, H., Yoon, H., Mupparapu, M., Li, J., 2020, “Artificial Intelligence for the Computer-aided Detection of Periapical Lesions in Cone-beam Computed Tomographic Images”, *Journal of Endodontics*, 46(7), 987-993.
- J47. Reisi Gahrooei, M., Yan, H.*, Paynabar, K., 2020, “Comments on: On Active Learning Methods for Manifold Data”, *TEST*, 29(1), 38-41.
- J48. Zhang, C., Yan, H.*, Lee, S., Shi, J., 2020, “Dynamic Multivariate Functional Data Modeling via Sparse Subspace Learning”, *Technometrics*, 63(3), 370-383. (**Win the Applied Track Best Paper Award in the in Data Mining Section of INFORMS**).
- J49. **Li, Z.**, Yan, H.*, Zhang, C., Tsung, F., 2020, “Long-Short Term Spatiotemporal Tensor Prediction for Passenger Flow Profile”, *IEEE Robotics and Automation Letters*, 5(4), 5010-5017. (**Win 2020 IEEE CASE Best Paper Award**).
- J50. Gahrooei, M. R., Yan, H.*, Paynabar, K., Shi, J., 2020, “Multiple Tensor-on-Tensor Regression: An Approach for Modeling Processes With Heterogeneous Sources of Data”, *Technometrics*, 63(2), 147-159.
- J51. **Kang, Y.**, Yan, H.*, Ju, F., 2020, “Performance Evaluation of Production Systems Using Real-Time Machine Degradation Signals”, *IEEE Transactions on Automation Science and Engineering*, 17(1), 273-283.

- J52. Yan, H.*, Paynabar, K., Shi, J., 2019, “AKM2D: An Adaptive Framework for Online Sensing and Anomaly Quantification”, *IISE Transactions*, 52(9), 1032-1046. (This paper is one of the four finalists of Best Student Paper Award in the Quality, Statistics, and Reliability Section of INFORMS).
- J53. Yan, H.*, Paynabar, K., Pacella, M., 2019, “Structured Point Cloud Data Analysis via Regularized Tensor Regression for Process Modeling and Optimization”, *Technometrics*, 61(3), 385-395.
- J54. Zhang, C., Yan, H.*, Lee, S., Shi, J., 2018, “Multiple Profiles Sensor-Based Monitoring and Anomaly Detection”, *Journal of Quality Technology*, 50(4), 344-362.
- J55. Yan, H.*, Paynabar, K., Shi, J., 2018, “Real-time Monitoring of High-Dimensional Functional Data Streams via Spatio-Temporal Smooth Sparse Decomposition”, *Technometrics*, 60(2), 181-197. (**Win Best Paper Award in the QSR Refereed Track of INFORMS 2015 and Best Student Paper Award in the Industrial and Systems Engineering Research Conference 2015**).
- J56. Zhang, C., Yan, H.*, Lee, S., Shi, J., 2018, “Weakly correlated profile monitoring based on sparse multi-channel functional principal component analysis”, *IISE Transactions*, 50(10), 878-891.
- J57. Yue, X., Yan, H.*, Park, J.G., Liang, Z., Shi, J., 2017, “A Wavelet-based Penalized Mixed-effects Decomposition for Multichannel Profile Detection of In-line Raman Spectroscopy”, *IEEE Transactions on Automation Science and Engineering*, 15(3), 1258-1271.
- J58. Yan, H.*, Paynabar, K., Shi, J., 2017, “Anomaly Detection in Images with Smooth Background Via Smooth-Sparse Decomposition”, *Technometrics*, 59(1), 102-114. (**Win Best Student Paper Award in Data Mining Section of INFORMS 2014**).
- J59. Yue, X., Wang, K., Yan, H.*, Park, J.G., Liang, Z., Zhang, C., Wang, B., Shi, J., 2017, “Generalized Wavelet Shrinkage of Inline Raman Spectroscopy for Quality Monitoring of Continuous Manufacturing of Carbon Nanotube Buckypaper”, *IEEE Transactions on Automation Science and Engineering*, 14(1), 196-207.
- J60. Mesnil, O., Yan, H.*, Ruzzene, M., Paynabar, K., Shi, J.[†], 2016, “Fast wavenumber measurement for accurate and automatic location and quantification of defect in composite”, *Structural Health Monitoring*, 15(2), 223-234.
- J61. Yan, H.*, Liu, K., Zhang, X., Shi, J., 2016, “Multiple Sensor Data Fusion for Degradation Modeling and Prognostics Under Multiple Operational Conditions”, *IEEE Transactions on Reliability*, 65(3), 1416-1426.
- J62. Yan, H.*, Paynabar, K., Shi, J., 2015, “Image-based Process Monitoring using Low-rank Tensor Decomposition”, *IEEE Transactions on Automation Science and Engineering*, 12(1), 216-227.

A2. Papers Submitted/In Revision

- J63. Hu, J., Yan, H.*, Li, Z., Zhang, C., “Tensor Completion for High-dimensional Data with Graph-structured and Weakly-dependent Sample Dimension”, *IEEE Transactions on Automation Science and Engineering*, in revision.

A3. Book Chapters

- B1. Yan, H.*, Li, Z., Zhao, X., Hu, J., 2024, “Sparse Decomposition Methods for Spatio-Temporal Anomaly Detection”, In *Multimodal and Tensor Data Analytics for Industrial Systems Improvement*, Springer, 185-206.

A4. Referred Conference Publication or Symposium Proceedings

- C1. **Chen, S.**, Moradi, M., Paynabar, K., Yan, H.*, 2026, “Flow Mismatching: Unsupervised Anomaly Detection via Velocity Discrepancies in Flow Matching Models”, *Advances in Neural Information Processing Systems (NeurIPS)*, 2026, accepted. arXiv:2605.23070.
- C2. **Feng, S.**, Yan, H.*, 2026, “Structurally Separated DAG Learning with Multi-Scale Normalized Closure”, *Advances in Neural Information Processing Systems (NeurIPS)*, 2026, accepted.
- C3. **Chen, S.**, Yan, H.*, 2026, “D-Convexity: A Unified Differentiable Convex Shape Prior via Quasi-Concavity for Data-driven Image Segmentation”, In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 34755-34764).
- C4. **Xu, B.**, Zou, Q., Yang, S., Yan, H.*, 2026, “Path-Coupled Bellman Flows for Distributional Reinforcement Learning”, *Proceedings of the 43rd International Conference on Machine Learning*, Seoul, South Korea, PMLR 306, 2026. arXiv:2605.08253.
- C5. **Xu, B.**, Yan, H.*, Reisi-Gahrooei, M., Ilbeigi, M., 2026, “Adaptive Sampling for Automated Post-Disaster Rapid Damage Assessment via Level-Set Cost-Aware Bayesian Optimization”, *Proceedings of the SIAM International Conference on Data Mining (SDM)*. arXiv:2608.02868.
- C6. **Huang, J.**, Xu, B., Liu, Y., Yan, H.*, 2025, “Multi-modal Generative Modeling of Event Sequences and Time Series for Solar PV Systems”, In 2025 IEEE 21st International Conference on Automation Science and Engineering (CASE) (pp. 345-350). (**Win 2025 IEEE CASE Best Conference Paper Award**).
- C7. **Zou, Q.**, Yan, H.*, 2025, “Probabilistic Kolmogorov-Arnold Networks via sparsified deep Gaussian processes with additive kernels”, In 2025 IEEE 21st International Conference on Automation Science and Engineering (CASE) (pp. 2889-2894).
- C8. **Li, Z.**, Yan, H.*, Zhang, C., Ketter, W., Tsung, F., 2023, “Tensor Dirichlet process multinomial mixture model with graphs for passenger trajectory clustering”, In Proceedings of the 6th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery (pp. 121-128).
- C9. **Wang, B.**[~], Tsung, F., Yan, H.*, 2022, “Attention-based Representation Learning for Time Series with Principal and Residual Space Monitoring”, In 2022 IEEE 18th International Conference on Automation Science and Engineering (CASE) (pp. 1833-1839).
- C10. **Zhao, X.**[~], Yan, H.*, Liu, Y., 2022, “Event Extraction for aviation accident reports through attention-based multi-label classification”, In AIAA AVIATION 2022 Forum (pp. 3831).

- C11. **Huang, J.**[~], Yan, H.^{*}, Li, J., Stewart, H. M., Setzer, F.C., 2021, “Combining Anatomical Constraints and Deep learning for 3-D CBCT Dental Image Multi-label Segmentation”, In 2021 IEEE 37th International Conference on Data Engineering (ICDE) (pp. 2750-2755).
- C12. **Zhao, X.**[~], Yan, H.^{*}, Liu, Y., 2021, “Hierarchical Tree-based Sequential Event Prediction with Application in the Aviation Accident Report”, In 2021 IEEE 37th International Conference on Data Engineering (ICDE) (pp. 1925-1930).
- C13. **Huang, J.**, Sergin, N. D., Dua, A., Tavakoli, EB, Yan, H.^{*}, Ren, F., Ju, F., 2020, “Edge Computing Accelerated Defect Classification Based on Deep Convolutional Neural Network With Application in Rolling Image Inspection”, In Proceedings of the ASME 2020 15th International Manufacturing Science and Engineering Conference (pp. V002T07A037).
- C14. **Li, Z.**^{*~}, Sergin, N. D., Yan, H.^{*}, Zhang, C., Tsung, F., 2020, “Tensor Completion for Weakly-dependent Data on Graph for Metro Passenger Flow Prediction”, *Thirty-Fourth AAAI Conference on Artificial Intelligence*, Vol 34(4), pp. 4804-4810.
- C15. **Zhao, X.**, Iquebal, A.S., Sun, H., Yan, H.^{*}, 2020, “Simultaneous Material Microstructure Classification and Discovery via Hidden Markov Modeling of Acoustic Emission Signals”, In Proceedings of the ASME 2020 15th International Manufacturing Science and Engineering Conference.
- C16. Yan, H.^{*}, Yeh, H., Sergin, N. D., 2019, “Image-based Process Monitoring via Adversarial Autoencoder with Applications to Rolling Defect Detection”, In 2019 IEEE 15th International Conference on Automation Science and Engineering (CASE) (pp. 311-316).
- C17. Yan, H.^{*}, Zhao, X., Hu, Z., Du, D., 2019, “Physics-based Deep Spatio-temporal Meta-modeling for Cardiac Electrical Conduction Simulation”, In 2019 IEEE 15th International Conference on Automation Science and Engineering (CASE) (pp. 152-157).
- C18. **Zhao, Y.**[~], Yan, H.^{*}, Holte, S., Mei, Y., 2019, “Rapid Detection of Hot-spot by Tensor Decomposition on Space and Circular Time with Application to weekly Gonorrhea data”, *The XIIIth International Workshop on Intelligent Statistical Quality Control*, Hong Kong.
- C19. **Zhao, X.**[~], Kang, Y., Yan, H.^{*}, Ju, F., 2019, “Semi-supervised Constrained Hidden Markov Model Using Multiple Sensors for Remaining Useful Life Prediction and Optimal Predictive Maintenance”, *Annual Conference of the Prognostics and Health Management Society (PHM)*, Vol 11.
- C20. **Zhao, X.**[~], Yan, H.^{*}, Li, J., Pang, Y., Liu, Y., 2019, “Spatio-temporal Anomaly Detection, Diagnostics, and Prediction of the Air-traffic Trajectory Deviation using the Convective Weather”, *Annual Conference of the Prognostics and Health Management Society (PHM)*, Vol 11.
- C21. **Kang, Y.**[~], **Yan, H.**, Ju, F., 2018, “Real-time production performance analysis using machine degradation signals: a two-machine case”, *2018 IEEE 14th International Conference on Automation Science and Engineering (CASE)*.
- C22. **Mesnil, O.**^{*~}, Yan, H.^{*}, Ruzzene, M., Paynabar, K., Shi, J., 2015, “Guided Wavefield Reconstruction from Sparse Measurements Using Compressed Sensing”, *International Workshop on Structural Health Monitoring*, Stanford, United States.

- C23. **Mesnil, O.*~**, **Yan, H.**, Ruzzene, M., Paynabar, K., Shi, J., 2014, “Frequency Domain Instantaneous Wavenumber Estimation for Damage Quantification in Layered Plate Structures”, *EWSHM - 7th European Workshop on Structural Health Monitoring*, Nantes, France.

C. INVITED TALKS

- T1. University of Cologne, Cologne Institute for Information Systems Research Seminar Series, “Knowledge-constrained learning and anomaly detection based on high-dimensional data”, Nov 2022
- T2. Gates Foundation Seminar, “Sequential Hotspot Detection in Spatio-temporal Systems ”, Oct 2022
- T3. Ebay Seminar Series, “Modern High-dimensional Anomaly Detection”, March 2022
- T4. University of Cologne, “Knowledge-constrained Deep Learning”, April 2022
- T5. Fred Hutchinson Cancer Research Center Seminar, “Sequential Hotspot detection in the Spatio-temporal Systems”, April 2021
- T6. Chinese Academy of Science Quality and Data Science Center Seminar Series in Academy of Mathematics and System Science, “Knowledge-infused Deep Learning for High-dimensional Data Monitoring and Modeling”, Jan 2021,
- T7. ASU 1st Machine Learning Day, West Campus, “Exploiting the Structure of High-Dimensional Data for Anomaly Detection”, Arizona State University, Mar 2019.
- T8. ASU Statistics Department, “Exploiting the Structure of High-Dimensional Data for Anomaly Detection”, Arizona State University, Sep 2018,
- T9. Virginia Tech INFORMS Student Chapter Seminar Series, “Exploiting the Structure of High-Dimensional Data for Anomaly Detection”, Apr 2018, Blacksburg, VA

V. PROFESSIONAL ACTIVITIES AND SERVICE

A. Professional Contributions and Training

A1. *Editor or Associated Editors Role*

- 2021 – present IISE Transaction Associate Editors
- 2023 – present IEEE Transaction on Automation Science and Engineering

A2. *Service to the Professional Organization:*

1. Chair Elect, INFORMS QSR, 2024
2. Council Member, INFORMS QSR, 2022 – 2024
3. Quality, Statistics, and Reliability 25th-year anniversary workshop, with more than 100 participants from around the world.
4. Organizing committee, 1st ICQSR conference
5. Chair, “INFORMS QSR Public Communication Committee”, 2022 - 2024
6. Student Introduction & Interaction Session committee, INFORMS QSR, 2022

7. Public Communication Subcommittees, INFORMS QSR, 2022-now
8. Membership growth program committee, INFORMS QSR, 2019-2020
9. Panel Reviewer for Department of Energy SBIR/STTR Program, 2022
10. Panel Reviewer for Department of Energy NEUP Program, 2022
11. Panel Reviewer for Department of Energy SETO Program, 2020
12. Quality, Statistics, and Reliability 1st workshop, INFORMS 2021
13. Quality, Statistics, and Reliability 20th-year anniversary workshop, with more than 100 participants from around the world.

A7. Professional Training

- 34th National Effective Teaching Institute Training Workshop, January 3-5, 2018, San Diego, CA
- ASU Career Proposal Workshop, 2019

VI. PERSONNEL: STUDENT SUPERVISION / MENTORING, TEACHING, DISSERTATION COMMITTEES, RESEARCHERS, AND OUTREACH

A. Teaching Activities

A1. Course Taught

- Instructor, IEE 640, Probability & Stochastic Processes, Fall 2023
- Instructor, IEE 577/CSE 598/IEE598 Data Science for System Decision Analytics, Spring 2018, Spring 2020, Spring 2021, Spring 2022, Spring 2023
- Instructor, IEE 605 Foundation of Information Systems: Spring 2019, Spring 2021
- Instructor, IEE 474 Quality Control, Fall 2021, Fall 2020, Fall 2019
- Instructor, IEE 572 Design and Analysis of Engineering Experiment, Fall 2017, Fall 2018

B. Mentoring Activities for Ph.D. students

1. Nurettin Dorukhan Sergin, Industrial Engineering, School of Computing and Augmented Intelligence, Graduated
 - Thesis: “Outlier-Aware Applications in High-Dimensional Industrial Systems”, 2017 Fall – 2021 Summer
 - Graduation Date: 07/2021
 - Current Position: Meta
 - Award: Best Paper Competition Finalist in 16th INFORMS Hybrid Workshop on Data Mining, and Decision Analytics applied track finalist, 2021, for the paper “Low-rank and Sparse Tensor Decomposition with Ridge Regularized Subspace Clustering for Metro Passenger Flow Modeling”.
2. Xinyu Zhao, Industrial Engineering, SCAI, 2018 Fall – 2022 Summer, Graduated

- Thesis: “Sequential Failure Event Modeling and Prediction in Aviation Accident Analysis”
 - Graduation Date: 07/2022
 - Current Position: Zillow
 - Award: QSR Best Referred Paper Award, 2020, for the paper “Adaptive Partially-Observed Sequential Change Point Detection with Multiple Failure Modes”.
3. Jiuyun Hu, Industrial Engineering program, 2020 Spring – 2024 Summer, Graduated
 - Award: IISE QCRE Best Student Paper Competition finalist for the paper “Tensor Completion for High-dimensional Data with Graph-structured and Weakly-dependent Sample Dimension”
 4. Jiayu Huang, Industrial Engineering program, SCAI, 2020 Fall – present
 5. Shuai Feng, Industrial Engineering program, SCAI, 2022 Fall – present
 6. Siqin Yang, Data Science, Analytics, and Engineering program, 2022 Fall – present (funded research assistant)
 7. Boyang Xu, Data Science, Analytics, and Engineering program, 2023 Fall – present (funded research assistant)
 8. Shengzhe Chen, Industrial Engineering program, SCAI, 2024 Fall – present (funded research assistant)
 9. Xianjian Xie, Data Science, Analytics, and Engineering program, SCAI, 2024 Fall – present

VII. RESEARCH SUPPORT

Bold: PI

- G1. Co-PI, “LEAP-HI: Sustainable Hydrogen Transport Pipeline Network with Learning-enabled Autonomous Risk Assessment Systems (LARAS), 09/01/2025 - 08/31/2029, team, Wenlong Zhang, Yiming Deng, Yongming Liu, Hanna Breetz, Total Amount: \$2,000,000 (20%)
- G2. PI, “Collaborative Research: Multi-Agent Adaptive Data Collection for Automated Post-Disaster Rapid Damage Assessment”, 09/01/2023 – 08/31/2026, team: **Mostafa Reisi, Mohammad Ilbeigi**, Total Amount: \$550,000 (30%)
- G3. PI, “Bayesian Optimization for Automatic Reactor Design Optimization”, team: **Andi Wang**, 10/01/2023 – 09/30/2026, Total Amount: \$1,000,000 (40%)
- G4. Co-I, “Novel threat detection methodology to detect HIV outbreaks in Washington”, National Institutions of Health R21, 9/1/2022 – 8/31/2024, team: **Sarah Holte, Roxanne P. Kerani**, Hao Yan, Yajun Mei, Total Amount: \$252,281.00 (75%)
- G5. Co-I, “AIDen: An AI-empowered detection and diagnosis system for jaw lesions using CBCT”, National Institutions of Health STTR Phase I, team: **Jing Li, Frank Setzer**, Hao Yan, Fleming Lure, Mel Mupparapu, 9/1/2022 – 8/31/2023. \$299,987 (20%)

- G6. Co-PI, “Knowledge-guided Automation for Integrity Management of Aging Pipelines (KAIMAP) for Hydrogen Transport”, Department of Transportation, team: **Yongming Liu**, Yiming Deng, Hao Yan, Total Amount \$844,726 (30%), 09/01/2022-08/31/2025
- G7. P.I., “Photovoltaic Plant Predictive Maintenance Optimization under Uncertainties Using Probabilistic Information Fusion”, Department of Energy (DOE), team: **Hao Yan**, Yongming Liu, Total Amount: \$1,130M (50%), 07/01/2022-06/31/2025
- G8. Co-PI, “Hybridizing Data and Model Driven Approaches for Proactive Production Control”, National Science Foundation (NSF), team: **Feng Ju**, Hao Yan, Total Amount: \$ 400,747 (30%), Period of Contract: 04/01/2020-03/31/2023
- G9. Co-PI, “Information Fusion for Real-Time National Air Transportation System Prognostics under Uncertainty”, NASA University Leadership Initiative, team: **Yongming Liu**, Aditi Chattopadhyay, Nancy Cooke, Jingrui He, Mary Niemczyk, Pingbo Tang, Lei Ying, Hao Yan, Sankaran Mahadevan, PK Menon, Barron Bichon, \$10M (10%), 01/01/2020 – 09/30/2023.
- G10. P.I., National Science Foundation (NSF), “ATD: Collaborative Research: Adaptive and Rapid Spatial-Temporal Threat Detection over Networks”, Amount: \$68,378 (100%), team: **Hao Yan**, **Yajun Mei**, Sarah Holte, Period of Contract: 9/1/2018-8/31/2021 (Completed),
- G11. P.I., Procter & Gamble Company, “Modeling Multi-Stage Manufacturing Processes and Related Problems.”, Amount: \$25000 (100%), Period of Contract: 7/1/2019-7/1/2020 (Completed), team: **Hao Yan**